A discrete spacetime of events in whole numbers: why clocks slow, why nothing outruns light, and why time has a direction only where something was measured

Alon Gonen

Independent researcher, Haifa, Israel.

Corresponding author(s). E-mail(s): alon@defounder.ai;

Abstract

In our view nature cannot be continuous: in quantum mechanics things come in whole pieces, and a measurement gives one outcome that cannot be unmade. We propose a discrete board of time events: Nodes of whole numbers, each speaking to six neighbours once per tick, one step per tick, and a gravity content setting each Node's rate; a click is a detector's reading of the board. Each tick a Node adds its neighbours' weighted numbers, subtracts its number of one tick before, divides once, remainder kept: every step runs back. From that rule we find one bound on everything on the board, no body outrunning light and, given a two-way reading by light, the known addition of speeds, clocks slowed near matter, moving clocks slowed in the known form at first order, light slowed near matter and, in form, bent and delayed, and a board time that no body reads. None of it gives time a direction; that comes only at a click: a detector takes one whole piece, records it, wipes the rest, one step per tick. With the record the board runs back through the click; without it the backward run fails where the wiping reached, no faster than one step per tick. On the board, the direction of time is the ledger of what was measured: a place, a speed, a size. One run shows it. We propose nature's arrow appears this way. Everything here is in the public repository and can be run by anyone.

Keywords: arrow of time, reversible rule, whole numbers, measurement, detectors, erasure

1 The problem and the answer

Physics today has no agreed answer to an old question [1–3]. Every basic law we know, one small piece of particle decay aside, works the same forward and backward. Yet everything we see goes one way. A cup breaks and never un-breaks.

One answer asks for a very special start [2–4]. Collapse models add a random jolt to the law, with a rate and a length put in [5], or a time set by gravity's own energy [6]. Decoherence places the arrow where the surroundings register a system [7, 8]; that the arrow is what can be recorded was argued and answered for quantum systems [9, 10]. The cost of erasing information has been counted [11–14]; that the loss sits where no copy of what was erased is kept is an old thesis [13], and new here is the exact count on a board, a grid of sites called Nodes, in three dimensions and in whole numbers. This paper too adds three things beside the step, what every Node does at each tick, the board's one beat: a toss, which is a weighted draw, a wiping, which overwrites numbers, and a write into the detector that took. What differs is this: the rule has no direction of time, the step stays exactly reversible in whole numbers, the toss runs back with its record, and the loss is counted exactly and bounded to one step per tick.

The claim: On a board of Nodes stepped by one whole-number rule, no step has a direction of time. A direction appears only where a detector we place, one Node or a few with a record, clicks (takes something from the numbers and records it) and wipes, and only for whoever has no record of what was wiped. The failure of the backward run then has a place, a size, the wiped count, and a spread never above one step per tick, each a whole number or a count of them.

If the world is built as such a board, this is where the direction of time in nature comes from: not in the law, but in the ledger of what was measured. We propose that it is so.

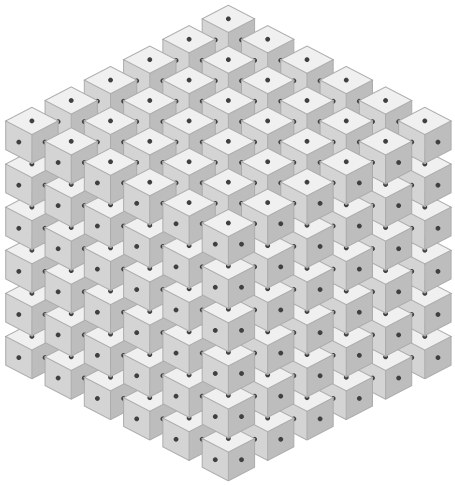
2 The board

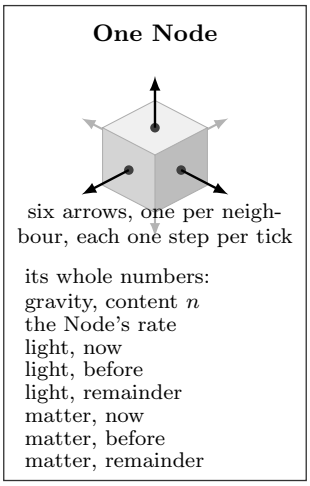
What we assume.

1. Nature is discrete: a board of events, Nodes at ticks, with nothing between them. It looks continuous not because it is, but because it is made of very many small parts and we measure from far: Section 5 puts the distance between Nodes under about 10^{-25} m, far below anything measured.
2. A Node reads its six neighbours (front and back, left and right, up and down) once per tick and writes only itself; nothing passes faster than one step per tick (the speed of light follows, below).
3. What we see is clicks, not the board.
4. Every number at a Node is a whole number; the step is exact, the remainder (what is left over from the division) kept.
5. Every Node does one simple act, the step of equation (1), the same for light and matter; each pattern of these carries two whole numbers, a top and a bottom; the kinds differ in how the two stand.
6. Light is the kind with the top equal to the bottom: it moves and does not turn in place. Matter is the kind with the top below the bottom: it turns in place even at rest.

The board: Nodes of whole numbers, six neighbours each, one bound

Each tick, every Node reads its six neighbours and writes only itself; nothing moves more than one step per tick; a long pattern of light moves at c, about 0.58 Node per tick.





$$w\,a_{\text{next}} + r' = \textstyle\sum_{\text{six}} R_j\,a_j + S\,a_{\text{now}} - w\,a_{\text{before}} + r, \quad 0 \le r' < w$$

the one rule at every Node, every tick, the remainder kept; every step can be undone.

Fig. 1 The board: a block of Nodes, six by six by six, each a cube with a dot on each face and from it a rod to the neighbour on that side; at the right one Node with its six arrows, one step per tick each, and its whole numbers; beneath, the rule, equation (1).

7. One kind, gravity, has a content at the Node, a count of units held there, fixed through a run, that sets the Node's rate, a whole number, q_0 at an empty Node, q_0 chosen, which sets how much of a swing it advances per tick (equation (7)), a swing being the numbers' going up, down and back.
8. The click, put in by hand: a detector, one Node or a few with a record, takes one whole piece, the smallest amount a detector can take, records it, and wipes the rest.

We conjecture a fourth kind, binding, that holds a body's Nodes together. This paper does not use it. It is what would let a detector be a body of the board, producing its own clicks and keeping its own record (Section 6). Everything else in this paper follows from these by the rule, or is counted on the board; where something is put in by hand, we say so.

Each Node has six neighbours. One event can touch another only through its six neighbours: a number held at a Node is read by its neighbours at the next tick, and by nobody else. Space and time on the board are read off these connections and off the clicks. One tick for every Node at once is what physics calls a preferred frame (Section 5, on what a body can read of it). Figure 1 draws the board.

A Node keeps three whole numbers for each of light and matter: now, before, and the remainder left over from the last division.

Each tick, every Node does the same thing at the same time. It reads the six numbers its neighbours hold now, each with a fixed weight; adds its own number now, with its own weight, which is zero where no gravity content stands; takes away its number from one tick before; and adds the old remainder. Then it divides the total by a fixed whole number w, keeping the quotient as its next number and the remainder. This is the rule's definition, taken from [15]; nothing in it is derived. As one line, with a_{next} the next number, a_j the number arriving from the neighbour j, R_j the weight of that neighbour, S the Node's own weight, a_{now} and a_{before} its two numbers, r the old remainder and r' the new one:

$$\boxed{w\,a_{\text{next}} + r' = \sum_{\text{six}} R_j\,a_j + S\,a_{\text{now}} - w\,a_{\text{before}} + r, \quad 0 \le r' < w.} \tag{1}$$

The right side is one whole number. The left side says how it is split: w times the next number, plus a remainder smaller than w. The numbers are whole numbers with sign; the quotient is the largest whole number not above the total over w, so the remainder lies in $0 \le r' < w$ for a total of either sign. A rule of the form "next = a function of the present numbers, minus the number before" runs back for any function [16, 17]. Ours is one such function, exact in whole numbers by the kept remainder. Boards of whole numbers have been proposed before [18–20], a spacetime of events as a partial order [21], and a world that is mathematics [22].

The kinds of pattern. A kind of pattern is chosen by two whole numbers, a top and a bottom number. With n units of gravity content at the Node the rule's definition says that each unit multiplies the rate by $(1 - 1/q_0)$, so the rate is $q_0\,(1 - 1/q_0)^n$.

The rule's definition gives the three weights at a Node from the two numbers and the rate. The divisor is $w = 6\,\text{bottom}\,q_0^2$. The rate squared over q_0 is the Node's reach toward each neighbour, and each neighbour's weight is twice the top number times that reach squared, $R_j = 2\,\text{top}\,(\text{rate}^2/q_0)^2$. The own weight is $S = 12\,(\text{bottom}\,q_0^2 - (\text{bottom} - \text{top})\,\text{rate}^2) - 6R_j$. At the empty rate, where the rate is q_0, the three are $6\,\text{bottom}\,q_0^2$, $2\,\text{top}\,q_0^2$ and 0: the own weight is zero wherever no gravity content stands. Two of the rule's whole numbers are rounded: the rate, the exact fraction $q_0\,(1-1/q_0)^n$ rounded half up once, and the reach, the rate squared over q_0, rounded half up once (rounded half up: the whole part of the quotient, plus one where the remainder is at least half the divisor); the weights R_j, S and w are then exact products of whole numbers. Equations (6) to (8) use the unrounded forms, which the rounding misses by at most a half in the rate and the reach. The pair has $1 \le \text{top} \le \text{bottom}$, q_0 is a whole number above 1, the content n a whole number at or above 0, and w is above 0. For light, top equals bottom, so the six weights add up to twice w: the Node follows the mean of its neighbours' numbers and nothing holds it in place. For matter, top is below bottom, so the six weights add up to less than twice w, and the shortfall turns the numbers in place even at rest.

Every step runs back. After the step, the numbers that were read are still on the board: each Node's old number is now its before number. Take w times the next number, add the new remainder, and subtract everything that was read. What is left is the old remainder minus w times the old before number. Divide that by w:

the quotient, its sign turned, is the old before number, and the remainder is the old remainder, because it is smaller than w. This follows from the rule alone.

The speed of light. A Node speaks only to its six neighbours, once per tick. So no number, no pattern and no news moves more than one step per tick: one bound for everything on the board. The wiping after a click is made to run at exactly that limit, outrunning light along a row and matching it along a body diagonal. Measured in a straight line, light's speed follows from equation (1) with light's weights, each a third of w and the own weight 0. A long pattern, many Nodes from one highest number to the next (crest to crest), then obeys "next plus before equals twice the mean of the six neighbours", whose speed squared for long patterns is one third: about 0.58 Node per tick. We write c for that speed.

The six neighbours sit as the six faces of a cube. The turnings and mirror images that leave the cube as it was form the cube's symmetry group, of order 48, the order being how many it holds. The step treats the six neighbours alike, so the step commutes with the group, remainders included: a symmetry then a step is a step then that symmetry. A long pattern's speed is therefore the same in every direction, because the group's only invariant of second degree, a number every symmetry leaves as it is, is the sphere's, the sum of the squares along the three axes; the cube enters at fourth degree (Section 5)[1], and only a short pattern feels the cube. Under the group the rate, the content, the count and the piece (Section 3) are scalars, unchanged by every symmetry; the board's momentum (Section 3) and what the click writes into the taker are vectors, turning with the axes; the wiped region and a body of one Node are invariant, each mapped onto itself. All of this follows from the rule and the group together.

Nature shows one top speed for light and for every signal, the same in every direction and the same whatever sent the light [23, 24]. The board has one bound for everything, one step per tick, and light at $c = 1/\sqrt{3}$ step per tick along a row, the same speed in length in every direction for long patterns (footnote 1). The bound and c are not one speed. In the board's frame two patterns of light meeting head-on close at $2c$; that is nobody's reading (Section 5). The wiping is carried Node to Node at the bound: a detector it reaches reads its arrival, the one bit the draw fixed, sooner than light from the click could have told it, $\sqrt{3}$ times along a row. News on the board moves at the bound; light, and every body of the matter kind, slower (Section 5). That is the title's "nothing": no body outruns light; news on the board may. Table 1 collects the paper's symbols, each with its one meaning.

3 The click

Two rules run the board: the step, which is reversible, and the click, which we declare, with a detector, a counting stretch, a toss and a write. A piece is one count in the detector's books, the count it keeps of pieces; on the board it is carried by a pattern of numbers moving from Node to Node.

[1] Counted in steps: one step per tick along a body diagonal, 0.816 along a face diagonal and 0.577 along a row; in length the same.

Table 1 The paper's symbols and named quantities, each with one meaning.

symbol	what it is	defined at	range or unit
w	the rule's divisor, 6 bottom q_0^2	equation (1)	whole number above 0
R_j, S	a neighbour's weight; the Node's own weight	equation (1)	whole numbers; $S = 0$ at no content
a_j	the number arriving from the neighbour j	equation (1)	whole number with sign
a_{now}, a_{before}, a_{next}	the Node's number now, one tick before, next	equation (1)	whole numbers with sign
r, r'	the old and the new remainder	equation (1)	$0 \le r' < w$
top, bottom	a pattern's pair	Section 2	$1 \le$ top $\le$ bottom; light: equal
q_0	the empty Node's rate, chosen	assumption 7	whole number above 1
n	the gravity content at a Node	Section 2	whole number, 0 or more
rate, reach	the Node's rate, $q_0\,(1 - 1/q_0)^n$ rounded; its reach, the rate squared over q_0, rounded	equation (6), Section 2	whole numbers, the rate q_0 when empty
c	the speed of long light on the empty board	Section 2	$1/\sqrt{3}$ Node per tick
swing, clock	a number's going up, down and back; a body's swings in so many ticks	assumption 7, Section 5	2π radians of turn; swings
piece, count, stretch	the smallest amount a detector takes; the books' count of pieces; the ticks a detector reads a piece over	assumption 8, Section 3	one whole; whole number; ticks
g, M, X_i, Y_i, turn_i	the detector's strength and scale, put in by hand; the detector i's two sums over a stretch; its turn at the stretch's end	equation (2)	whole numbers; radians
W_i, W_0, U, P_i	the detector's weight; the no-click weight; what is left of a piece; the detector's chance	equations (2), (3)	0 to 1
t, $N(t)$, L	ticks since the click; the wiped Nodes within t steps; a tube's width	equation (4)	whole numbers; Nodes
F_{kj}, P_a	the flow into the Node k from its neighbour j, from their numbers now_k, before_k; the board's momentum along the axis a	equation (5)	whole numbers, the equation's unit
x, y, z	a Node's place on the run's board	Section 4	Nodes
body	a pattern of numbers on the board, an atom say	Section 5	
turn, turn_0	the Node's turn per tick; the same at the empty rate	equation (7)	radians; cosine 0 to 1
s, Φ	the slope of the swings in the content, $2\tan(\text{turn}_0/2)/\text{turn}_0$; the known potential of gravity	Section 5	1 within 5×10^{-16} for nature's clocks (Section 5); not mapped to n
$c(n)$, ℓ, Δn	light's speed at a Node of content n; a path's length; the difference of n across a ray	equation (8)	Node per tick; Nodes; whole number
p, v	a moving pattern's part of a swing between Nodes; its middle's speed	equation (9)	radians; Node per tick, below c
c_m	the speed standing for c in the matter kind's own form	Section 5	its square $\text{turn}_0 \cot(\text{turn}_0)/3$; below c
v_A, v_B, u	two bodies' speeds; what the first reads of the second by two-way light	Section 5	Node per tick
A, T, $\mathbf{d}$, d_a	a pattern's height; a count of ticks; a unit direction and its parts on the axes	Section 5	whole numbers; $\sum_a d_a^2 = 1$
E, δv, $E_{\text{QG},2}$	the light's energy; the change of its speed; the scale of a slowing growing as the energy squared	Section 5	GeV; Node per tick; GeV
ℓ_0, λ	the distance between neighbouring Nodes; the light's length, crest to crest	Section 5	metres
$\hbar$, ν, τ	the constant turning a rate of swinging into an energy; swings per second; the tick's length	Section 5	GeV s; per second; seconds

In the run each detector was an atom of two Nodes with two states, low and high, counting over a fixed stretch of ticks. Over the stretch the detector i adds up the numbers at its Nodes, each multiplied by one of two fixed up-and-down number sequences of its own, giving two sums, X_i and Y_i. The size of the two sums together, times a declared strength g, over a fixed scale M, is the detector's turn, an angle measured as a part of one full swing of the detector's own numbers; this is put in by

hand, and the detector's weight W_i in the toss is the square of the sine of that turn:

$$\text{turn}_i = \frac{g\sqrt{X_i^2 + Y_i^2}}{M}, \qquad W_i = \sin^2(\text{turn}_i). \tag{2}$$

In the run the weight is computed as a ratio of two whole numbers: the detector keeps one whole number for each of its two states, the whole of a fixed unit in the low state at first and 0 in the high; at a stretch's end the two sums of equation (2) give the turn, the root of equation (2) taken once, as the largest whole number whose square is at most $X_i^2 + Y_i^2$; the pair of numbers is then turned by that turn in whole numbers, and the weight is the high state's number squared over the sum of the two squares; the sine of equation (2) is that ratio's form, and the toss is drawn from the ratios outside the board. At the end of a stretch one toss is made over every detector that read the piece. The no-click weight W_0 comes from the books: what is left of one piece, U, one at the first stretch, lowered by the weights of every earlier stretch in which no click came, less the weights now, never below zero; each detector's chance P_i is its weight over the sum of all the weights, the no-click one included:

$$W_0 = \max\Big(\min(U, 1) - \sum_{\text{all}} W,\ 0\Big), \qquad P_i = \frac{W_i}{W_0 + \sum_{\text{all}} W}. \tag{3}$$

This is a counter on whole numbers, reported as fractions, read against the detector's two sequences. No toss is made while the count stands at zero.

If the toss says take, the click happens. One whole piece is taken: the count goes from 1 to 0, the atom from its low state to its high, and its own numbers are rewritten. Then the wiping begins at the Node that took the piece and reaches one step farther each tick, the board's limit. At each wiped Node the rule is given a value chosen so that the rule itself writes zero, given at one neighbour for two ticks: it writes the piece's now to zero at the first tick and its before to zero at the second; the remainder stays as the rule computes it.

The detector's record holds four kinds of entry: the numbers its own Nodes held before they were rewritten, the two given values at every wiped Node, how far the wiping has reached, and one line saying that a piece was taken, and when.

With the record, every click runs back; the argument is short. Each act of a click is a rewriting or a wiping, and each has one way back: put the old number back from the record. Between the acts the board steps by the rule, which has one way back (Section 2). A chain of steps that each have one way back has one way back: undo the last, then the one before, down to the first. So the whole run, click included, runs backward number for number. The weights are read from the gravity content, which nothing here moves, so the same weights serve forward and back.

Without the wiped numbers in the record, the board cannot be run back; one may ask where it fails first. The backward run has the board's numbers as they stand and the record, nothing else; whoever holds the start and the toss's seed, the number that fixes its draws, holds a copy of what was wiped and can rerun the board forward; the claim is about running back from the board as it stands, with that copy and the wiping's record both gone. The first step back starts where the wiping last reached.

The division there needs the number that was wiped out, and reads the replacement instead. So the result is wrong at those Nodes. Each further step back, a Node next to a wrong Node reads a wrong number and may go wrong itself (wrong numbers can also cancel), and no Node farther than one step can be touched in one tick; so the wrong set grows by at most one step on each side per tick. The rule allows no more.

What the wiping overwrote, counted. Two whole numbers per wiped Node, its now and its before, and two at the taker. The wiped set is a ball counted in steps: from the taking Node the wiping reaches the Nodes at t steps at its t-th tick, and a step being one connection, the layer at t steps holds $4t^2+2$ Nodes for $t \geq 1$ and one Node at $t = 0$, so by the rule's one step per tick the ball within t steps holds, counting the layers up:

$$N(t) = \frac{4t^3 + 6t^2 + 8t + 3}{3}, \qquad 1,\ 7,\ 25,\ 63,\ 129,\ \ldots \tag{4}$$

In a tube, a board whose sides hold zero, L Nodes wide, the ball fills the width and then runs along the tube, naming $2L^2$ Nodes and $4L^2$ values per tick.

What the step keeps. The step keeps one more thing, exactly without the remainder where the rate is the same at every Node: a momentum of the board along each axis, and we shall need it at the click. Write now_k and before_k for the two numbers of the Node k, $k+a$ and $k-a$ for its neighbours along the axis a, and $F_{kj} = \mathrm{now}_k\,\mathrm{before}_j - \mathrm{before}_k\,\mathrm{now}_j$ for the flow into the Node k from its neighbour j, a cross-difference of the kind called a Wronskian; by equation (1) the board's momentum along a, the flow in from one side less the flow in from the other, summed over every Node, is kept without the remainder where the rate is the same at every Node and no number stands at the board's two ends along that axis (an end, holding zero, exchanges momentum with the board; in the run the light never reached them):

$$P_a = \sum_k \big(F_{k,k-a} - F_{k,k+a}\big). \tag{5}$$

Why is it kept? Where the rate is the same at every Node, equation (1) without the remainder reads $\mathrm{next}_k + \mathrm{before}_k = \sum_j (R_j/w)\,a_j + (S/w)\,\mathrm{now}_k$, with the same weight from a Node to its neighbour as from the neighbour back; put next for now and now for before in equation (5): the weighted terms cancel in pairs, one each way along the axis, and the terms in before give P_a back. With the remainder the step changes P_a by less than $2\sum_k |\mathrm{now}_{k-a} - \mathrm{now}_{k+a}|$, each remainder below w, and the changes add up over ticks. The books keep the count of pieces; that is the books' rule, not the step's. At a click the taker is given the whole piece (in the run one piece lay as two halves of light, Section 4): its count, and the momentum of what it took as read at its Nodes, written as a turn of the taker's numbers that grows from Node to Node along the axis; the turn carries that momentum where the taker can hold it, else the largest it holds, a quarter swing: the click writes into the taker four fifths of what the half that reached it carried: its two Nodes read 0.86 of it and the quarter swing's cap left 0.80 (Section 4); the wiping of the other half's tail is where the board's momentum is lost. Because the piece is one whole, the count, kept by the books (above), falls to zero at the click, and no toss is made while it stands at zero: a second detector's chance is

Table 2 One step at one Node with the run's $q_0 = 6{,}000$, at the worked pairs.

kind	neighbours' sum × weight	$w\times$ before	total	next	remainder
light, pair 1 and 1	864,000,000	216,000,000	648,000,000	3	0
matter, pair 2 and 3	1,728,000,000	648,000,000	1,080,000,000	1	432,000,000

zero from that tick, wherever it stands. The count's fall reaches every detector at the same tick through the books, outside the board; the books' acts, the toss's sum and the count's fall, are the only things here not bound to one step per tick. Two detectors that read one piece are bound by it, and we say they are tied; on the board nothing carries the tie. The tie is the count's conservation in whole pieces.

4 The run

The run's board. The run's q_0 was 6,000. The run's light ran at the pair 6,000 and 6,000 and its atoms at 1 and 1,299; Table 2 and the swings are worked at the pair 1 and 1 for light and 2 and 3 for matter. For light's pair 1 and 1 at the empty rate the divisor w is 6 times 6,000 squared, 216,000,000, each neighbour's weight is 2 times 6,000 squared, 72,000,000, a third of w, and the own weight is 0; for matter's pair 2 and 3, w is 648,000,000, each neighbour's weight 144,000,000, the own weight 0. One step at a Node whose six neighbours hold 3, 3, 2, 2, 1, 1 and which holds 4 now, 1 before and the remainder 0 is in Table 2. The step back, for light: 216,000,000 times 3, plus 0, minus 864,000,000 is $-216{,}000{,}000$; divided by 216,000,000, -1 with 0 over; the sign turned, 1 before and the remainder 0.

The board was fixed, 400 by 8 by 8 Nodes, 25,600 in all, its ends and sides holding zero, a tube eight Nodes wide. Two detectors, atoms of two Nodes, sat at (175, 4, 4) and (176, 4, 4) and at (223, 4, 4) and (224, 4, 4) (x, y, z, counted in Nodes), each counting over 48 ticks, with the strength 40 and the toss's seed 25. The detectors' two sequences were a cosine and a sine at light's turn per tick for eight Nodes crest to crest, equation (9) with top equal to bottom, the cosine 5,414/6,000, whole numbers at the scale $M = 4{,}096$ advanced each tick by the rule next $= (2 \cdot 5{,}414 \cdot \text{now})/6{,}000 - \text{before}$, the quotient rounded half up; the turn of equation (2) was taken in units of $1/q_0$, and the toss was one draw per stretch from a fixed arithmetic sequence begun at the toss's seed. At the click the taker's low part was set to no piece and its high part to one, its number the unit, the numbers at its two Nodes turned by the light's angle at the stretch's end, the angle whose tangent is Y_i/X_i; the wiping gave each wiped Node the chosen value at its neighbour one step nearer the taker, for two ticks. Two patterns of light were laid at the middle, one running each way, eight Nodes from crest to crest, heights 21 and 20, booked as one piece. The run was 172 ticks.

At tick 48 the weights were 0.080 and 0.109, and the no-click weight W_0 was 0.811; neither detector clicked. At tick 96 the weights, 0.536 and 0.678, exceeded the 0.811 left, so W_0 was 0, and the detector at the larger x, at (223, 4, 4), took the piece with the chance 0.56. The count fell to zero, and from then no toss was made: the other detector, at (175, 4, 4), still read the piece's other half, its weight 0.039 at tick 144, and could not click.

Over seeds. The same world was run 400 times more, the toss's seed the only change, 1 to 400, and then 400 times with the near detector, the one at the smaller x, alone, and 400 with the far one alone. With both present, the near detector took the piece in 174 runs and the far one in 226: the frequencies 0.435 ± 0.025 and 0.565 ± 0.025, against 0.438 and 0.562 from equation (3) applied stretch by stretch, at ticks 48 and 96, with the weights of a run in which no click came; of the 324 clicks at tick 96, 182 were the far detector's, 0.562, against the one run's chance 0.56. With both present, no run had two clicks and none had no click, as equation (3) and the books force. The far detector alone took the piece in 326 runs of 400, 0.815 ± 0.019, 0.812 by equation (3), against 0.565 with the near one present; the near one alone, 263 of 400, 0.658 ± 0.024, 0.658 by the same equation, against 0.435 with the far one present. The cause is in equation (3): at tick 96 the two weights together, 1.21, exceed the 0.81 left of the piece, so the no-click weight is 0 and the whole piece is split between the two in the proportion of their weights; alone, each keeps its own weight and the rest goes to no click. So on this board, with the click as put in, a detector's presence changes another's rate within one stretch, across 48 Nodes that light crosses in 83 ticks, through the books and not the board; the counts show it. The weights of equation (2) are put in by hand; the frequencies are counts of runs on the board.

The wiping named 6, 18, 38, 64, 90, 110, 122, 127, 128 Nodes at ticks 97 to 105, then 128 per tick, the full tube's $2L^2$ at $L = 8$: 9,279 Nodes and 18,560 values by the run's end, two per Node and two at the taker; 18,176 written by tick 172, 16,111 of them not zero before they were overwritten; the last two ticks' 384 named and not yet written. At the run's end the light's other half, its crest at Node 119, still lay mostly outside the ball, whose edge at Node 148 had just reached its tail.

Three checks, each a count on the board. First, the board with the click against the same board without it: they differed only inside the wiped ball, from tick 96 on, never outside. Second, backward with the record: every number matched, 171 down to 0. Third, backward with the wiping's record emptied and the rest kept: the first wrong numbers came at tick 171, where the wiping had just reached, and spread never faster than one step per tick in any direction; the wrong set's two ends along the board moved 60 and 8 Nodes in 171 ticks, and 13,673 of the 25,600 Nodes were wrong at tick 0.

The swings on the board. At the pair 2 and 3 and $n = 600$ the rate is 5,429, 0.905 of $q_0 = 6{,}000$; by equation (7) a Node then swings 0.900 as fast as at the empty rate, and the first-order slope of Section 5 is 1.06. One such Node, its six neighbours holding its own number, counted 722.5 swings in 6,000 ticks at 600 units of content against 803.0 at none, 0.900; the Node's numbers were laid with a height of 10^6, the two rows' below with 10^7 (the largest numbers 1.0×10^6 at rest, 9.1×10^6 on the moving row and 1.0×10^7 on the light row). With the pattern's neighbouring Nodes an eighth of a swing apart, equation (9) gives 0.1967 Node per tick and 0.916 of the swings at rest, and the square of the speed that stands for c in the matter kind's own form (Section 5) is 0.251, 0.50 against light's 0.58, and the fourth-order factor of Section 5 is 1.69. The worked pair, its turn near 1, is far from nature's (Section 5). On a row of 400 Nodes a moving middle counted 147.5 swings in 1,200 ticks against 160.5 at rest, 0.92, at 0.1965 Node per tick against 0.1967 by the form. A long pattern of light on a row at 600 units of content moved at 0.818 of its empty-board speed, against 0.819

by equation (8). Equation (5) summed over the board: the light's P_x was 4,280 at tick 96 before the click (3,480 at tick 0; 872 at tick 172 without a click, the remainder's drift). The piece lay as two halves of light running opposite ways. Run alone, the half the taker took carried 39,272 at tick 96 (37,952 at the start) and the other −38,584 (−36,640); the two on one board, 3,480 at the start, are not their sum 1,312: the flow of equation (5) is a product of two numbers, so where the halves were laid together their cross terms, 2,168, add to the sum, and the step keeps them; at tick 96 the rest of the gap, about 1,400, is the remainder's drift on the three boards. A pattern has a size on the board: at each Node w times the squares of its two numbers, less its own weight times now times before, less now times the weighted sum of its six neighbours' before numbers, all over twice the reach squared; summed over its Nodes the step keeps it in its linear form, the remainder beside, and by it the books count one piece; this is the rule's definition, not derived here. The taker was written 31,560, 0.80 of its half's momentum. The reading at the taker's two Nodes, the terms of equation (5) there divided by the share of one piece's size those two Nodes held, so that it stands for one whole piece, was 33,812, 0.86 of its half's momentum, and the quarter swing's cap, the angle $1/q_0$ taken 9,424 times, held back 2,252 more. The same two-Node reading across the tube, 34,349 and 13,418, is not the light's (the whole light carried −1,124 and 0 across the tube), and nothing was written across, the atom being one Node wide there. The wiping then zeroed that half: at tick 120 the light stood at −26,406 and the taker at 31,560, together 5,154 against 2,936 without a click, so the click with the zeroing of its half keeps P_a to within 2,218 of the board without a click. By the run's end the wiping's edge had cut into the other half's tail, leaving sharp ends whose terms add to the sum, and the light's P_x stood at −46,774, light and both atoms together −14,842 against 4,280 before: that cut is where the board's momentum was lost. The step keeps P_a but for the remainder's drift; the click four fifths; the wiping's cut into the other half does not.

5 What time is on the board

A body's clock is how many swings it completes in so many ticks. A body, a pattern of numbers on the board, an atom say, cannot see the board's tick; it counts its own swings and its own clicks, and that is its time. One body's count differs from another's in pace for two reasons, and in direction for a third.

Where it stands. With n units of gravity content the rule of Section 2 gives the rate:

$$\text{rate} = q_0\,(1 - 1/q_0)^n. \tag{6}$$

From the rule it follows that a Node whose six neighbours hold its own number steps, by equation (1), to next plus before equals now times twice the cosine of the turn, the angle its numbers turn through in one tick, a part of one swing; the neighbours' weight cancels in that cosine. In the words top, bottom and rate, with q_0 the empty rate, the rule gives the turn:

$$\cos(\text{turn}) = 1 - \Big(1 - \frac{\text{top}}{\text{bottom}}\Big)\Big(\frac{\text{rate}}{q_0}\Big)^2. \tag{7}$$

Section 4 counts it on the board; this turn is the Node's, per tick, and the detector's turn of equation (2) is another angle; angles are in radians, a full swing 2π. Equations (7) to (10) are the rule's linear form, the remainder's term $(r - r')/w$ dropped; that term is below one unit of the Node's number per tick, so a pattern of height A follows them to within about $1/A$ per tick, T/A over T ticks; at heights of a few units the rule leaves them (a Node holding 1 may fall to 0 in one step). Nature's clocks lower in gravity run slow: the measured shift of light climbing in gravity [25] gives, to first order in the potential Φ, the known measure of how deep in gravity a clock sits, the rate $1 + \Phi/c^2$ [26]. Ours, from equation (7) at small content, is linear in the content as the known one is in the potential: the swings fall as $1 - s\,n/q_0$ with the slope $s = 2\tan(\mathrm{turn}_0/2)/\mathrm{turn}_0$, turn_0 the turn at the empty rate, nearing 1 as top nears bottom. The forms agree at first order, and s is 1 to the order counted below, at the tick's bound.

Light near matter. Near matter light still follows the mean of its neighbours, but with the share $(\mathrm{rate}/q_0)^4/3$ each and the rest on itself; by equation (1) a long pattern of light then moves at the speed $c(n)$ at a Node of content n:

$$c(n) = c\left(\frac{\mathrm{rate}}{q_0}\right)^2 = c\,(1 - 1/q_0)^{2n}, \tag{8}$$

about $c\exp(-2n/q_0)$, while the swings of equation (7) fall as $1 - s\,n/q_0$. In other words light slows twice as strongly as the rate, $2 : s$ against the clock; the known ratio is $2 : 1$. The two powers are the definition's, the rate to the fourth in R_j against the second in S (Section 2), so the board meets the ratio and nothing here derives it; s is 1 for nature's clocks, as shown below; we counted the speed on a row (Section 4). Both slowings, light's and the clock's, are read in the board's own ticks and Nodes (Section 6). Where the content changes slowly from Node to Node a long pattern follows the ray's law, that a path bends toward where it moves slower: the ray turns toward the content by $2/q_0$ times the difference Δn of n across the ray, per Node, and over a path of length ℓ it is late by $(2/(cq_0))\int n\,d\ell$, to first order in n/q_0, the forms of the bending and the delay seen near the Sun [27, 28] (Section 6). The rate of equation (6) is above zero at every n, so by the form no clock stops and light leaves every Node, only slower: the form has no horizon. In whole numbers the reach rounds to zero once the rate falls below $\sqrt{q_0/2}$, at a content near $(q_0/2)\ln(2q_0)$: such a Node reads nothing from its neighbours, so no pattern enters it and the wiping's write, given through a neighbour, cannot reach it; its neighbours still read it, and its own number is not frozen: a matter Node swings on alone at the slow turn equation (7) gives at that rate; a light Node, its turn 0 there, changes by the same amount each tick, next $= 2\,\mathrm{now} - \mathrm{before}$.

How it moves. A moving body is a pattern of numbers handed on from Node to Node, a fixed distance each tick, and each Node it reaches is a fixed part p of a swing behind the one it left (seen from far, the pattern looks like a wave), so part of each swing goes to the passing: a Node the pattern passes through swings faster than it would with the body at rest, but the number at the body's middle swings less, by the part the handing takes. By equation (1), such a Node's turn, the middle's speed v and the middle's turn per tick, its swings per tick being this over 2π, as Section 4's counts

have it, are:

$$\cos(\text{turn}) = \frac{\text{top}}{\text{bottom}}\Big(1 - \frac{1-\cos p}{3}\Big), \qquad v = \frac{d\,\text{turn}}{dp}, \qquad \text{middle's turn} = \text{turn} - p\,v. \tag{9}$$

Nature's moving clocks run slow: the measured light of fast-moving atoms [29] gives the rate $(1-v^2/c^2)^{1/2}$ [30], to first order $1-v^2/(2c^2)$. Ours, from equation (9) at small speed, is $1 - v^2/(2c_m^2)$ with $c_m^2 = \text{turn}_0 \cot(\text{turn}_0)/3$, c_m the speed that stands for c in the matter kind's own form, light's $c^2 = 1/3$ only as top nears bottom. This is the branch of long patterns, p small; for short ones the middle counts more than at rest: the two counts cross at p about two at the worked pair, where the pattern moves at $0.32c$, and nearer $p = 0$ as top nears bottom, where it moves near c; the result is the branch below that crossing. The form is the known one and the coefficient is the pattern's own, c_m in place of c; how near c_m comes to c for a clock of nature, the tick's bound tells below. The board's form departs at fourth order by $2\,\text{turn}_0/\sin(2\,\text{turn}_0)$, 1 as top nears bottom. From equation (9) a body of the matter kind never moves faster than c: its speed squared stays below $1/3$ for every top below bottom. A body at the speed v_A that sends light to another, at v_B, and counts the returns by its own swings, with light as its ruler, reads the other's speed as $u = (v_B - v_A)/(1 - v_A v_B/c^2)$, below c, exactly at every pair, for long flashes on the empty board, the factor of the body's own count cancelling in the ratio; at light's c, not the pattern's own c_m. On the board light runs at c; the first body sends two flashes along their common line, a gap apart; the second, moving at v_B, meets them a gap $(c - v_A)/(c - v_B)$ times as long, and the two returns reach the first a gap $(c + v_B)/(c + v_A)$ times as long as that. Its own count scales both gaps by one factor, which cancels in their ratio. It reads the distance as c times half the time between sending and return, and the time as their mean, so it reads the speed $c\,(\text{return gap} - \text{sending gap})/(\text{return gap} + \text{sending gap}) = (v_B - v_A)/(1 - v_A v_B/c^2)$, below c whenever both are. That the count at the moving middle is what the body's clicks read, we propose (Section 7). If so, a moving clock counts fewer seconds.

How short the tick is. By equation (1) with light's weights a long pattern moves at c; a short one, with the part p of a swing from Node to Node, follows equation (9) with top equal to bottom and moves a little slower, along an axis by

$$v = c\,\Big(1 - \frac{p^2}{12} + \dots\Big), \tag{10}$$

along a face diagonal by $c\,(1 - p^2/48 + \dots)$, and not at all along a body diagonal, where the turn is $p/\sqrt{3}$ at every p. Why do the three differ? The group's first invariant beyond the sphere's, $\sum_a d_a^2 = 1$, is $\sum_a d_a^4$ for the unit direction $\mathbf{d}$ with d_a its part on the axis a, and a short pattern's slowing goes as $(3\sum_a d_a^4 - 1)/24$ times p^2: $1/12$ on an axis, $1/48$ on a face diagonal, 0 on a body diagonal, $1/30$ averaged over directions; this follows from the rule. Nature's light shows no such slowing. The light of one distant burst, seen up to the energy $E = 1$ TeV, arrived with its speed v changed by a part $|\delta v/v|$ below 3.8×10^{-16}; written as $\frac{3}{2}\,(E/E_{\text{QG},2})^2$, with $E_{\text{QG},2}$ the scale in which such a slowing is written (not an energy at which it would equal the speed), this puts $E_{\text{QG},2}$ above 6.3×10^{10} GeV at 95 per cent confidence for light slowed, the

board's case, the burst's own lags fitted beside it [31], one bound of this kind and not the tightest. A second burst, seen to 31 GeV, gives 1.3×10^{11} GeV [32], which would halve the lengths below; a third, seen to 13 TeV [33], gives 6.9×10^{11} GeV at 95 per cent confidence [34], eleven times the first. The three stand 64°, 71° and 130° apart on the sky (from the positions reported in [31–33]); the body diagonals stand 70.5°, 109.5° or, at the two ends of one diagonal, 180° apart, so the pair at 130° fits no two of them within ten degrees, five for each, and no turn of the board hides all three near them; bounds of this kind, first sought in the bursts [35], are reviewed in [36]. Write $p = 2\pi\,\ell_0/\lambda$, with ℓ_0 the distance between neighbouring Nodes and λ the light's length from one crest to the next, 1.24×10^{-18} m at 1 TeV; $\hbar$ is the constant that turns a rate of swinging into an energy, $E = 2\pi\hbar\nu$ for ν swings per second. Then ℓ_0 is shorter than $3\sqrt{2}\,\hbar c/E_{\mathrm{QG},2} = 1.3 \times 10^{-26}$ m if that light ran along an axis of the board, and shorter than 1.1×10^{-25} m in every direction more than five degrees from a body diagonal (the bound weakens toward it, eightfold at 5°, fortyfold at 1°, and is absent on it); the direction of the burst on the board we do not know. Light moves 0.577 of that distance per tick, so the tick, the discreteness of time, is shorter than $\sqrt{6}\,\hbar/E_{\mathrm{QG},2} = 2.6 \times 10^{-35}$ s along an axis and than 2.2×10^{-34} s in every direction more than five degrees from a body diagonal; with the three bursts 64° to 130° apart, at least one lies more than ten degrees from every body diagonal however the board is turned, so 2.2×10^{-34} s holds in every way the board may lie (the tightest bound would give 2×10^{-36} s on an axis). A clock of nature turns through $\nu\tau$ of a swing per tick, $2\pi\nu\tau$ in radians, with τ the tick's length in seconds; taking a body's swing rate at rest as its rest energy over $2\pi\hbar$, an identification we add, the inner clock proposed before any board [37], the electron's 1.24×10^{20} swings per second and τ under 2.2×10^{-34} s give turn_0 under 1.8×10^{-13}, so top over bottom, $\cos(\mathrm{turn}_0)$, lies within 2×10^{-26} of 1, s within 3×10^{-27} of 1 and c_m^2 within 4×10^{-27} of 1/3; nature's matter kind then has a bottom above 6×10^{25}; the proton, 1,836 times faster, within 6×10^{-20}, 10^{-20} and 2×10^{-20}. For every clock up to a uranium atom's rest rate the board's coefficients are the known ones, $1 - v^2/(2c^2)$ and 2 : 1, if the middle's count is the body's time: $s = 1$ within 5×10^{-16} and the moving clock's coefficient within 2×10^{-15} (c_m within 10^{-15} of c); $s - 1$ is 3×10^{-27} at the electron's rate, 10^{-20} at the proton's, 3×10^{-17} at the iron nucleus's, 2×10^{-16} at a caesium atom's, 5×10^{-16} at uranium's, and the moving clock's coefficient departs by four times as much, $\mathrm{turn}_0^2/3$. A click is written in one tick; the stretch over which a detector reads a piece is a count it declares, a whole number of half swings, or many swings: over such a stretch the size of the two sums of equation (2) no longer depends on where in the swing it began.

A prediction. If light of higher energy is ever seen to arrive later than light of lower energy from the same distant source, beyond what the source itself does, the board says how. The delay grows as the square of the energy, equation (10), and never the other way: on the board no light is faster than c. The delay differs by direction in the cube's pattern, set in the board's axes, which the data would place on the sky, 1/12 along an axis, 1/48 along a face diagonal, none along a body diagonal; and its size fixes the tick. Of the delay only the part that grows with the distance to the source is the board's; a source's own lag does not grow so, and sources at several distances tell the two apart. That part, found at the confidence the field uses, 95 per cent, growing in proportion to the energy, or with light arriving early, or the same in every direction once sources in enough directions are compared, would refute the board. The numbers above are a bound from a measurement, not a prediction; the form of the slowing is the prediction.

6 What is open

The detector is placed by us, its reset after a click and the keeping of its record ours too, outside the board; in nature the detector is a body on the board and keeps the rule, so nothing it does outruns light, and what it does in detail is not built here. The click's weights of equation (2), put in by hand, are not the piece's shares: they do not add up to one over the detectors, so where together they exceed what is left of the piece, equation (3) splits it between them in proportion, and a detector's presence changes another's rate within a stretch (Section 4): the one toss at the stretch's end sums every detector's weight in equation (3), by the books, outside the board, as the count's fall is (Section 3). So the click as put in has a global part, the books, which read and act on every detector at once, and that is where the far effects come from; the bound of one step per tick is the board's and holds for everything on it, numbers, patterns and the wiping, not for the click's books, which stand outside it. Nature's detectors too share one piece, but each takes it at its own rate whether or not the other stands; the board's do not. The mend is a weight that is the detector's share of the piece, the detectors' shares and the no-click share adding up to one; we have not done it here, and Section 4's counts over seeds show the fault as it stands.

The wiping carries the click's one bit at the bound, $\sqrt{3}\,c$ along a row, to a detector in its path; nature shows no signal above c; whether a body of the board, not a detector we place, could read the wiping's arrival we have not asked of the board. The bending and the delay of light near matter are not counted on the board. How a mass's content falls off from Node to Node is not set here, so whether a mass of nature shuts light in is not shown. What a moving body reads with a ruler of its own, the two-way sameness that was measured [24], and what it reads of light from one side, and what a body standing in the content reads of light with its own clock and its own ruler, are not shown here; the one-side reading depends on the matter pair, and we do not settle it here.

Which pattern of the board a clock of nature is we do not say; a single pattern heavier than about 10^{-17} kg would, at the tick's bound, turn past a quarter swing

Table 3 Time on the board against what is seen.

	what is seen	on the board	where they agree, and where not
the top speed	one top speed for light and for every signal, the same in every direction and whatever the source [23]	one bound for every number and signal, one step per tick; light at $c^2 = 1/3$ of it for long patterns, the same in every direction; the wiping at the bound (Section 2)	the bound in form; no body of the matter kind outruns light, derived; one body's two-way reading of another the known addition at c, exact; light below the bound along a row, where the wiping outruns it by $\sqrt{3}$; a moving body's own ruler [24] and the one-way reading (Section 7)
the clock near matter	the rate $1 + \Phi/c^2$; light climbing in gravity [25, 26]	equations (6) and (7); to first order $1 - s\,n/q_0$	linear in n as the known is in Φ, n not mapped to Φ here; row 3's $2:1$ follows the definition's powers of the rate, no map needed
light near matter	light bends and is delayed near the Sun [27, 28]	equation (8), $c(n) = c\,(\text{rate}/q_0)^2$; the bending and the delay at first order	the form at first order; light's slowing $2:s$ against the clock's, the known $2:1$ by the definition's powers of the rate; $s = 1$ within 5×10^{-16} (Section 5)
the moving clock	the rate $(1 - v^2/c^2)^{1/2}$; fast-moving atoms [29, 30]	equation (9); to first order $1 - v^2/(2c_m^2)$	first order at c_m, the speed that stands for c in the matter kind's own form; $s = 1$ within 5×10^{-16} and $c_m = c$ within 10^{-15} (Section 5); the fourth order departs by a known factor; the identification proposed ours alone; proposed for nature
the direction of time	a measurement cannot be unmade; no agreed formula	the record of what was measured: a place and a size, the wiped count $N(t)$ of equation (4), from which the failure spreads	

per tick, which equation (7) forbids: at the empty rate $\cos(\text{turn}_0)$ is top over bottom, between 0 and 1; whether such a body is one pattern or many is not shown here. Whether a body keeps its own record we do not know. A classical piece is as exclusive; whether the tie gives the correlations that in nature tell entanglement from a shared piece we have not tested.

7 Conclusion

The claim: A board of Nodes, each speaking only with its six neighbours once per tick and stepped by one whole-number rule, has no direction of time: every step runs back exactly. The direction appears only where detectors we place click: the wiping removes the piece's numbers, and for anyone without a record of them (in the run, our log) the backward run fails where the wiping reached; the failure has a place, a speed never above one step per tick, and a size, a count of whole numbers. By the rule no body forgets.

Why a board like this agrees with what is seen. Nothing below was fitted: the eight assumptions and the rule's definition of Section 2, its weights with their powers of the rate, are taken as given, and what follows comes from them and from what each item names as put in or assumed, nothing else (Table 3).

1. One bound for everything. A Node speaks only to its six neighbours, once per tick, so nothing on the board, light, matter or news, moves more than one step per tick (the books' count is not on the board); equation (1) with light's weights gives long patterns of light the speed squared $c^2 = 1/3$ of that bound, the same

in every direction [derived from the rule's linear form]. A body of the matter kind never outruns light. From equation (9) a body's speed squared stays below one third for every top below bottom: $v^2 < 1/3 = c^2$ [derived from the rule's linear form]. By two-way light no body reads another as faster than light. A body that sends light to another and counts the returns by its own swings, with light as its ruler, reads the other's speed as $u = (v_B - v_A)/(1 - v_A v_B/c^2)$, the known addition of speeds at light's c, below c at every pair [light at c from the rule; the two-way reading put in by hand; exact].

2. Clocks run slow near matter. Each unit of gravity content at a Node multiplies its rate by $(1 - 1/q_0)$, equation (6) [the rule's definition]; by equation (7) the swings then fall as $1 - s\,n/q_0$ with $s = 2\tan(\mathrm{turn}_0/2)/\mathrm{turn}_0$: linear in the content, as the known rate $1 + \Phi/c^2$ is linear in the potential [linear in the content as any smooth rule is at first order; s derived from the rule's linear form, the remainder's term dropped; counted on the board (Section 4); the content not mapped to the potential].
3. Moving clocks run slow. A moving pattern spends part of each swing on the passing, equation (9), and its middle completes $1 - v^2/(2c_m^2)$ of the swings at rest, with $c_m^2 = \mathrm{turn}_0 \cot(\mathrm{turn}_0)/3$, c_m the speed that stands for c in the matter kind's own form: the known form at first order, c_m nearing c as top nears bottom; at fourth order the board departs by $2\,\mathrm{turn}_0/\sin(2\,\mathrm{turn}_0)$ [derived from the rule's linear form, the remainder's term dropped; counted on the board (Section 4); $s = 1$ within 5×10^{-16} and $c_m = c$ within 10^{-15} (Section 5); that the middle's count is the body's time, proposed].
4. Time has a direction only where something was measured. The click is put in: the weight the square of the sine of the turn, equation (2), the chance the weight over the sum, equation (3) [put in by hand; run; counted over seeds]. The wiping then reaches one step farther each tick, two whole numbers per wiped Node, $N(t)$ Nodes within t steps by equation (4), $2L^2$ Nodes per tick in a full tube of width L [the wiping's reach, one step per tick, declared with the click; $N(t)$ counted; the failure's spread derived from the rule; shown in the run]. Without the record the backward run fails where the wiping last reached, and the failure spreads never above one step per tick each way [derived from the rule; shown in the run]: the arrow's place, its size, the wiped count $N(t)$, and its spread, never above one step per tick, each a whole number or a count of them.

No body reads the one time. The board has one tick for every Node, a preferred frame, but no body reads it; two bodies disagree in pace for two reasons, where they stand (item 2) and how they move (item 3), and in direction for a third, what they measured (item 4) [from items 1 to 3; proposed].

The same equations give more. By equation (1) a long pattern of light moves at $c(n)$ of equation (8), slower near matter by twice the rate's share [derived from the rule's linear form, the remainder's term dropped; counted on a row (Section 4)]. Where the content changes slowly, light bends toward matter and is late, to first order, $2 : s$ against the clock, the known $2 : 1$ as top nears bottom [the form derived; the ratio the definition's; not counted]. A short pattern of light feels the cube, equation (10), and the light of distant bursts, seen to 13 TeV [33] and showing no such slowing [31, 32, 34],

puts the tick of time under about 2×10^{-34} s in every way the board may lie, three bursts in three directions leaving no diagonal to hide in, and under 2.6×10^{-35} s on an axis (2×10^{-36} s by the tightest bound) [derived from the rule's linear form; the bound measured; the form a prediction].

Why do we propose it? We assume the world is a very large number of events, each speaking only with its neighbours and following one rule, for us the step of equation (1). Economy is our reason, not evidence. On a board built so, the loss sits where the click wiped, and that nature's arrow sits there is the proposal.

What is different here. Earlier pictures of a discrete world [17–20] run many rules, or state the programme and leave the rule open; here one rule, equation (1), for light and matter, with gravity's content in its weights, is written in full, run (Section 4) and set against what is seen, and every statement above carries its status. Collapse models put in a rate and a length of nature by hand [5], or set the time by gravity's own energy [6]; here the chance of each outcome is a count of the board's numbers, equation (3), one draw, one whole piece written back, and what is put in by hand is the click, assumption 8, with the strength g and the scale M of equation (2) and the toss of equation (3): the detector's numbers and its draw, not a rate or a length of nature. Gravity is not a field added beside the rule: it is a kind of the same rule, its content a count held at the Node, laid by us and moved by nothing in this paper; the count enters the step through the weights R_j and S of Section 2, and one rate, equation (6), composed unit by unit, slows the clock, equation (7), and light, equation (8), in the ratio $2 : s$, the definition's powers of the rate. A body would be a pattern that holds itself together on the board; that needs the binding kind conjectured in Section 2.

A click picks one outcome. The step keeps the board's momentum of equation (5), exact at one rate without the remainder and bounded each tick with it; the click writes into the taker four fifths of what the half that reached it carried, its two Nodes reading 0.86 of it and the quarter swing's cap leaving 0.80, and the wiping's cut into the other half is where the board loses momentum (Section 4); the count of pieces the books keep. A click gives one whole piece to one taker, so a second detector's chance for it is zero from that tick, wherever it stands: that is the count's doing; in the run the count fell to zero and no second click came. The other outcomes tied to the chosen one are cancelled by the count in whole pieces. The tie itself, two detectors bound by one piece, is what we call entanglement on the board.

Statements and Declarations

Funding. None.

Competing interests. None.

Data availability. The run's world (one_piece_3d_fixed.json, with its mode file beside it) and its report (report_e4.json) are in the public repository at https://github.com/Closer24/Universe24 under papers/spacetime_events/run3d_fixed, with run_e4.py, which regenerates the record from the world and the mode file, and ensemble_seeds.py (under run3d_fixed/ensemble), which reruns the world over the seeds 1 to 400 with both detectors and with each alone and writes the counts of Section 4 into ensemble_seeds.json; under papers/spacetime_events/tools are

run_clocks.py, which counts the swings and the light row of Sections 4 and 5 into run_clocks.json, run_momentum.py, which reads the momentum at the click of Section 4 into run_momentum.json, and the scripts that rebuild this paper and rerun its checks. To repeat the run: load the world, run it forward to the run's end with the record, then backward with the record, then backward with the wiped-out numbers removed. Run on the archived version 2.0.0, these scripts rewrite the record, the figure's data and the four row files number for number; only the program's stamp, the status and the times differ. The rule that turns the toss's seed into its draws, a multiplier and an increment, is declared in the world file beside the seed.

Code availability. The program that runs the board is archived with its own version DOI [38].

Author contributions. A.G. did all the work.

Use of AI tools. The author used AI tools in preparing this manuscript, under the author's direction and review; the author takes full responsibility for its content.

References

[1] Loschmidt, J.: Über den Zustand des Wärmegleichgewichtes eines Systems von Körpern mit Rücksicht auf die Schwerkraft. I. Sitzungsberichte der Kaiserlichen Akademie der Wissenschaften, Mathematisch-Naturwissenschaftliche Classe **73**, 128–142 (1876). https://www.zobodat.at/publikation_series.php?id=7341

[2] Boltzmann, L.: Über die Beziehung zwischen dem zweiten Hauptsatze der mechanischen Wärmetheorie und der Wahrscheinlichkeitsrechnung respektive den Sätzen über das Wärmegleichgewicht. Sitzungsberichte der Kaiserlichen Akademie der Wissenschaften, Mathematisch-Naturwissenschaftliche Classe **76**, 373–435 (1877). Reprinted in Wissenschaftliche Abhandlungen, vol. II, pp. 164–223, Cambridge University Press, Cambridge (2012), https://doi.org/10.1017/CBO9781139381437.011

[3] Boltzmann, L.: Entgegnung auf die wärmetheoretischen Betrachtungen des Hrn. E. Zermelo. Annalen der Physik **293**(4), 773–784 (1896) https://doi.org/10.1002/andp.18962930414

[4] Rovelli, C.: Where was past low-entropy? Entropy **21**(5), 466 (2019) https://doi.org/10.3390/e21050466

[5] Ghirardi, G., Rimini, A., Weber, T.: Unified dynamics for microscopic and macroscopic systems. Physical Review D **34**(2), 470–491 (1986) https://doi.org/10.1103/PhysRevD.34.470

[6] Penrose, R.: On gravity's role in quantum state reduction. General Relativity and Gravitation **28**(5), 581–600 (1996) https://doi.org/10.1007/BF02105068

[7] Zurek, W.H.: Decoherence, einselection, and the quantum origins of the classical. Reviews of Modern Physics **75**(3), 715–775 (2003) https://doi.org/10.1103/RevModPhys.75.715

[8] Joos, E., Zeh, H.D.: The emergence of classical properties through interaction with the environment. Zeitschrift für Physik B Condensed Matter **59**(2), 223–243 (1985) https://doi.org/10.1007/BF01725541

[9] Maccone, L.: Quantum solution to the arrow-of-time dilemma. Physical Review Letters **103**(8), 080401 (2009) https://doi.org/10.1103/PhysRevLett.103.080401

[10] Jennings, D., Rudolph, T.: Entanglement and the thermodynamic arrow of time. Physical Review E **81**(6), 061130 (2010) https://doi.org/10.1103/PhysRevE.81.061130

[11] Landauer, R.: Irreversibility and heat generation in the computing process. IBM Journal of Research and Development **5**(3), 183–191 (1961) https://doi.org/10.1147/rd.53.0183

[12] Bennett, C.H.: Logical reversibility of computation. IBM Journal of Research and Development **17**(6), 525–532 (1973) https://doi.org/10.1147/rd.176.0525

[13] Bennett, C.H.: The thermodynamics of computation—a review. International Journal of Theoretical Physics **21**(12), 905–940 (1982) https://doi.org/10.1007/BF02084158

[14] Zurek, W.H.: Algorithmic randomness and physical entropy. Physical Review A **40**(8), 4731–4751 (1989) https://doi.org/10.1103/PhysRevA.40.4731

[15] Gonen, A.: Universe24: the implementation of one integer rule on a cubic lattice, read through clicks. Zenodo, concept record (2026). https://doi.org/10.5281/zenodo.23190113

[16] Toffoli, T., Margolus, N.: Invertible cellular automata: a review. Physica D **45**(1–3), 229–253 (1990) https://doi.org/10.1016/0167-2789(90)90185-R

[17] Fredkin, E.: Digital mechanics: an informational process based on reversible universal cellular automata. Physica D **45**(1–3), 254–270 (1990) https://doi.org/10.1016/0167-2789(90)90186-S

[18] Zuse, K.: Rechnender Raum. Vieweg, Braunschweig (1969). https://doi.org/10.1007/978-3-663-02723-2 . English translation: Calculating Space, MIT Technical Translation AZT-70-164-GEMIT, MIT Project MAC, Cambridge, 1970

[19] Wolfram, S.: A New Kind of Science. Wolfram Media, Champaign (2002). https://www.wolframscience.com/nks/

[20] 't Hooft, G.: The Cellular Automaton Interpretation of Quantum Mechanics.

Fundamental Theories of Physics, vol. 185. Springer, Cham (2016). https://doi.org/10.1007/978-3-319-41285-6

[21] Bombelli, L., Lee, J., Meyer, D., Sorkin, R.D.: Space-time as a causal set. Physical Review Letters **59**(5), 521–524 (1987) https://doi.org/10.1103/PhysRevLett.59.521

[22] Tegmark, M.: The mathematical universe. Foundations of Physics **38**(2), 101–150 (2008) https://doi.org/10.1007/s10701-007-9186-9

[23] Brecher, K.: Is the speed of light independent of the velocity of the source? Physical Review Letters **39**(17), 1051–1054 (1977) https://doi.org/10.1103/PhysRevLett.39.1051

[24] Michelson, A.A., Morley, E.W.: On the relative motion of the Earth and the luminiferous ether. American Journal of Science **34**(203), 333–345 (1887) https://doi.org/10.2475/ajs.s3-34.203.333

[25] Pound, R.V., Rebka, G.A.: Apparent weight of photons. Physical Review Letters **4**(7), 337–341 (1960) https://doi.org/10.1103/PhysRevLett.4.337

[26] Einstein, A.: Über den Einfluß der Schwerkraft auf die Ausbreitung des Lichtes. Annalen der Physik **340**(10), 898–908 (1911) https://doi.org/10.1002/andp.19113401005

[27] Dyson, F.W., Eddington, A.S., Davidson, C.: A determination of the deflection of light by the Sun's gravitational field, from observations made at the total eclipse of May 29, 1919. Philosophical Transactions of the Royal Society of London. Series A **220**(571–581), 291–333 (1920) https://doi.org/10.1098/rsta.1920.0009

[28] Shapiro, I.I.: Fourth test of general relativity. Physical Review Letters **13**(26), 789–791 (1964) https://doi.org/10.1103/PhysRevLett.13.789

[29] Ives, H.E., Stilwell, G.R.: An experimental study of the rate of a moving atomic clock. Journal of the Optical Society of America **28**(7), 215–226 (1938) https://doi.org/10.1364/JOSA.28.000215

[30] Einstein, A.: Zur Elektrodynamik bewegter Körper. Annalen der Physik **322**(10), 891–921 (1905) https://doi.org/10.1002/andp.19053221004

[31] Acciari, V.A., Ansoldi, S., Antonelli, L.A., Arbet Engels, A., Baack, D., Babić, A., Banerjee, B., Barres de Almeida, U., Barrio, J.A., Becerra González, J., *et al.*: Bounds on Lorentz invariance violation from MAGIC observation of GRB 190114C. Physical Review Letters **125**(2), 021301 (2020) https://doi.org/10.1103/PhysRevLett.125.021301

[32] Vasileiou, V., Jacholkowska, A., Piron, F., Bolmont, J., Couturier, C., Granot, J.,

Stecker, F.W., Cohen-Tanugi, J., Longo, F.: Constraints on Lorentz invariance violation from Fermi-Large Area Telescope observations of gamma-ray bursts. Physical Review D **87**(12), 122001 (2013) https://doi.org/10.1103/PhysRevD.87.122001

[33] LHAASO Collaboration: Very high-energy gamma-ray emission beyond 10 TeV from GRB 221009A. Science Advances **9**(46) (2023) https://doi.org/10.1126/sciadv.adj2778

[34] Cao, Z., *et al.*: Stringent tests of Lorentz invariance violation from LHAASO observations of GRB 221009A. Physical Review Letters **133**(7), 071501 (2024) https://doi.org/10.1103/PhysRevLett.133.071501

[35] Amelino-Camelia, G., Ellis, J., Mavromatos, N.E., Nanopoulos, D.V., Sarkar, S.: Tests of quantum gravity from observations of γ-ray bursts. Nature **393**(6687), 763–765 (1998) https://doi.org/10.1038/31647

[36] Mattingly, D.: Modern tests of Lorentz invariance. Living Reviews in Relativity **8**, 5 (2005) https://doi.org/10.12942/lrr-2005-5

[37] de Broglie, L.: Recherches sur la théorie des quanta. Annales de Physique **10**(3), 22–128 (1925) https://doi.org/10.1051/anphys/192510030022

[38] Gonen, A.: Universe24: the implementation of one integer rule on a cubic lattice, read through clicks. Zenodo, version 2.0.0 (2026). https://doi.org/10.5281/zenodo.23289515